

Feature Engineering: Mathematical Foundations

Derivation of the 163-Dimensional State Representation from Raw
Options Chain Data for the GBT Directional Classifier

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Contents

1	Motivation and Design Principles	2
2	Underlying Price Dynamics	2
2.1	Spot Price Normalization	2
2.2	Multi-Horizon Log Returns	2
2.3	Volatility-Adjusted Returns	3
3	Options Greek Surface	3
3.1	Black-Scholes Greek Derivations	3
3.2	Higher-Order Greeks	4
3.3	Net Dealer Exposure Aggregates	4
4	Dealer Gamma Exposure and Spot Dynamics	4
4.1	GEX as a Regime Indicator	4
4.2	Dominant Gamma Strike and Confluence	5
5	Initial Balance Geometry	5
5.1	Theoretical Motivation	5
5.2	Normalized Distance Features	5
5.3	Multi-Day Lookback Structure	6
6	Implied Volatility Surface Parameterization	6
6.1	ATM Implied Volatility and Term Structure	6
6.2	Volatility Skew	7
6.3	Smile Slope	7
7	Cross-Ticker Feature Construction	7
7.1	VIX as a Volatility Regime Signal	7
7.2	TLT as a Risk-Off Signal	7

1 Motivation and Design Principles

The GBT classifier operates on tabular data derived from a non-stationary financial time series. Unlike neural architectures that can implicitly learn feature representations, Gradient Boosted Trees partition the input space through axis-aligned splits. Predictive performance therefore depends critically on the quality of the pre-computed feature set: splits on raw, non-stationary price levels generalize poorly across the multi-year walk-forward training horizon, while splits on standardized, economically interpretable quantities yield robust cross-window performance.

This document formalizes the mathematical derivations behind each feature category, with particular attention to the transformations that induce approximate stationarity in the resulting feature distributions. The canonical target of these transformations is to produce a feature vector $\mathbf{x}_t \in \mathbb{R}^{163}$ whose marginal distributions are approximately time-stationary i.e., $\mathbf{x}_t \stackrel{d}{\approx} \mathbf{x}_{t+\Delta}$ for all reasonable Δ so that splits learned on historical training windows transfer to held-out test windows without systematic bias.

2 Underlying Price Dynamics

2.1 Spot Price Normalization

The raw spot price S_t is highly non-stationary (integrated of order 1). Direct use of S_t as a feature would create splits of the form $S_t > K$ that are meaningless once the underlying trends away from the training range. We therefore replace S_t with a rolling z-score:

$$x_t^{\text{spot}} = \frac{S_t - \mu_{S,20}}{\sigma_{S,20}}, \quad \mu_{S,20} = \frac{1}{20} \sum_{k=1}^{20} S_{t-k\Delta}, \quad \sigma_{S,20} = \sqrt{\frac{1}{20} \sum_{k=1}^{20} (S_{t-k\Delta} - \mu_{S,20})^2} \quad (1)$$

where the lookback window is 20 bars of 5 minutes each (100 minutes). This feature is bounded in practice to $[-3, +3]$ by the normal quantile and is $\mathcal{O}(1)$ regardless of the absolute price level of the underlying.

2.2 Multi-Horizon Log Returns

Log returns are the natural stationary transform of a geometric Brownian Motion:

$$r_{t,\Delta} = \log S_t - \log S_{t-\Delta} \approx \frac{S_t - S_{t-\Delta}}{S_{t-\Delta}}, \quad \Delta \in \{1, 5, 15, 30\} \text{ min} \quad (2)$$

Under the Black-Scholes model $S_t = S_0 \exp(\mu t + \sigma W_t)$, the log return $r_{t,\Delta} \sim \mathcal{N}(\mu\Delta, \sigma^2\Delta)$ is strictly stationary and normally distributed. In practice, equity returns exhibit excess kurtosis and volatility clustering, motivating the volatility-adjusted transform in Section 2.3.

2.3 Volatility-Adjusted Returns

The conditional variance of intraday returns is dominated by the implied volatility of the ATM 0DTE option. Scaling returns by the expected volatility over the horizon Δ normalizes the distribution across different volatility regimes:

$$\tilde{r}_{t,\Delta} = \frac{r_{t,\Delta}}{\sigma_t^{\text{ATM}} \sqrt{\Delta / (252 \times 390)}} \quad (3)$$

where σ_t^{ATM} is the annualized ATM implied volatility from the 0DTE chain, and 252×390 converts annual volatility to the per-minute scale.

Proposition 1 (Approximate Stationarity of Volatility-Adjusted Returns). *Under a stochastic volatility model $dS_t = S_t(\mu dt + \sigma_t dW_t)$, the quantity $\tilde{r}_{t,\Delta}$ has bounded variance conditional on σ_t^{ATM} :*

$$\text{Var}(\tilde{r}_{t,\Delta} \mid \sigma_t^{\text{ATM}}) \approx 1 + \mathcal{O}\left(\frac{\text{Var}(\sigma_t)}{\sigma_t^2}\right) \quad (4)$$

In periods of low volatility-of-volatility, $\tilde{r}_{t,\Delta}$ is approximately unit-variance, providing consistent scaling for GBT split thresholds across all market regimes.

Sketch. By Itô's lemma, $r_{t,\Delta} \approx \sigma_t W_\Delta$ to first order. If σ_t^{ATM} is a good predictor of σ_t , then $\tilde{r}_{t,\Delta} \approx W_\Delta / \sqrt{\Delta}$, which is a standard normal increment with unit variance. The error term arises from vol-of-vol effects. $\square$

3 Options Greek Surface

3.1 Black-Scholes Greek Derivations

Under the risk-neutral measure, the price of a European call with strike K , expiry T , at current time t with spot S_t , risk-free rate r , dividend yield q , and implied volatility σ is:

$$C(S_t, K, T - t, r, q, \sigma) = S_t e^{-q(T-t)} \Phi(d_1) - K e^{-r(T-t)} \Phi(d_2) \quad (5)$$

$$d_1 = \frac{\log(S_t/K) + (r - q + \sigma^2/2)(T - t)}{\sigma \sqrt{T - t}}, \quad d_2 = d_1 - \sigma \sqrt{T - t} \quad (6)$$

where Φ is the standard normal CDF. The primary Greeks are:

$$\Delta = \frac{\partial C}{\partial S} = e^{-q(T-t)} \Phi(d_1) \quad (7)$$

$$\Gamma = \frac{\partial^2 C}{\partial S^2} = \frac{e^{-q(T-t)} \phi(d_1)}{S_t \sigma \sqrt{T - t}} \quad (8)$$

$$\theta = \frac{\partial C}{\partial t} = -\frac{S_t \sigma e^{-q(T-t)} \phi(d_1)}{2\sqrt{T - t}} + q S_t e^{-q(T-t)} \Phi(d_1) - r K e^{-r(T-t)} \Phi(d_2) \quad (9)$$

$$\mathcal{V} = \frac{\partial C}{\partial \sigma} = S_t e^{-q(T-t)} \phi(d_1) \sqrt{T - t} \quad (10)$$

where ϕ is the standard normal PDF. Near expiration ($T - t \rightarrow 0$), $\Gamma \rightarrow \infty$ for ATM options, which motivates the sign-preserving log transform (Equation ??) applied to aggregate Greek

sums.

3.2 Higher-Order Greeks

The system includes two third-order Greeks of particular relevance to 0DTE trading:

Vanna ($\partial\Delta/\partial\sigma = \partial^2 C/\partial S\partial\sigma$) quantifies how the delta of a position changes as implied volatility moves. When market makers are long vanna, a VIX spike forces them to buy additional spot hedge as their deltas increase, creating a self-reinforcing upward price move:

$$\text{Vanna}_i = \frac{\partial^2 C_i}{\partial S \partial \sigma} = -e^{-q\tau} \phi(d_1) \frac{d_2}{\sigma}, \quad \tau = T - t \quad (11)$$

Charm ($\partial\Delta/\partial t$) captures delta decay over time, particularly relevant for 0DTE positions where theta-delta interaction accelerates:

$$\text{Charm}_i = \frac{\partial^2 C_i}{\partial S \partial t} = qe^{-q\tau} \Phi(d_1) - e^{-q\tau} \phi(d_1) \frac{2(r - q)\tau - d_2\sigma\sqrt{\tau}}{2\tau\sigma\sqrt{\tau}} \quad (12)$$

3.3 Net Dealer Exposure Aggregates

Market-maker hedging pressure on the spot market is approximated by aggregating Greeks across the options chain weighted by open interest:

$$\mathcal{G}_t^{\text{net}}(\text{Greek}) = \sum_{i=1}^{N_{\text{chain}}} \text{OI}_i \cdot \text{Greek}_i \cdot \text{Mult} \quad (13)$$

where $\text{Mult} = 100$ for SPX (each contract controls 100 index units). Since OI values can reach 10^5 – 10^6 and gamma values peak near ATM-0DTE at 10^{-3} – 10^{-2} , products can easily exceed 10^4 . The log transform of Equation ?? is essential here.

4 Dealer Gamma Exposure and Spot Dynamics

4.1 GEX as a Regime Indicator

The relationship between aggregate GEX and realized spot volatility is well-documented in the market microstructure literature. Let GEX_t be computed as in Definition 3 of the Features document. The key regime classification is:

Definition 1 (GEX Regime).

$$\text{GEX}_t > 0 \Rightarrow \text{“Pinned” regime: dealers sell rallies, buy dips} \quad (14)$$

$$\text{GEX}_t < 0 \Rightarrow \text{“Trending” regime: dealers buy rallies, sell dips} \quad (15)$$

In the pinned regime, realized volatility is suppressed relative to implied volatility, and mean-reversion strategies around the max-gamma strike are expected to outperform. In the trending regime, momentum strategies are favored. The GBT learns this regime-conditional behavior from the data without explicit hard-coding.

4.2 Dominant Gamma Strike and Confluence

The dominant gamma strike K_Γ^* is the strike with the largest aggregate notional gamma exposure:

$$K_\Gamma^* = \arg \max_{K \in \mathcal{K}} \sum_{\substack{i \in \text{chain} \\ K_i = K}} \Gamma_i \cdot \text{OI}_i \cdot \text{Mult} \cdot S_t^2 \quad (16)$$

The spatial proximity of spot to this level is encoded via a Gaussian RBF kernel with bandwidth proportional to the spot price:

$$\Phi_t^\Gamma = \exp\left(-\frac{(S_t - K_\Gamma^*)^2}{2(b \cdot S_t)^2}\right), \quad b = 0.005 \text{ (50 bps)} \quad (17)$$

The bandwidth $b = 0.005$ corresponds to approximately 25 SPX points at the current price level, chosen to capture the typical pin-radius observed empirically in 0DTE market maker hedging behavior.

5 Initial Balance Geometry

5.1 Theoretical Motivation

The Initial Balance (IB) represents the price range established during the first 60 minutes of trading (09:30–10:30 ET). Market Profile theory [1] postulates that the IB serves as a reference frame for the remainder of the session: price tends to either remain within (“inside day”) or break out of (“trending day”) this initial range, with distinct autocorrelation structures in each case.

Formally, for trading day d , define:

$$\text{IB}_{\text{high}}^{(d)} = \sup_{\tau \in \mathcal{T}^{(d)}} S_\tau, \quad \text{IB}_{\text{low}}^{(d)} = \inf_{\tau \in \mathcal{T}^{(d)}} S_\tau, \quad \mathcal{T}^{(d)} = [09:30, 10:30]_d \quad (18)$$

The midpoint and width of the IB are:

$$\text{IB}_{\text{mid}}^{(d)} = \frac{\text{IB}_{\text{high}}^{(d)} + \text{IB}_{\text{low}}^{(d)}}{2}, \quad \text{IB}_{\text{width}}^{(d)} = \text{IB}_{\text{high}}^{(d)} - \text{IB}_{\text{low}}^{(d)} \quad (19)$$

5.2 Normalized Distance Features

The distance from spot to each IB boundary is normalized by a basis-point scaling factor $\nu = 0.005$ (50 bps of spot price) to produce features that are commensurate across different absolute price levels:

$$\delta_{\text{high}}^{(d)} = \text{clip}\left(\frac{S_t - \text{IB}_{\text{high}}^{(d)}}{\nu \cdot S_t}, -10, 10\right) \quad (20)$$

$$\delta_{\text{low}}^{(d)} = \text{clip}\left(\frac{S_t - \text{IB}_{\text{low}}^{(d)}}{\nu \cdot S_t}, -10, 10\right) \quad (21)$$

When $\delta_{\text{high}}^{(d)} > 0$, spot is above the IB high; when $\delta_{\text{high}}^{(d)} < 0$, spot is within or below. The clip at ± 10 bounds the features to $\pm 5\%$ of spot price.

5.3 Multi-Day Lookback Structure

Historical IB levels from prior trading days $d \in \{-1, -2, -3, -4, -5\}$ serve as multi-day support/resistance reference frames. For each prior day d , three features are computed:

$$\mathcal{F}_{\text{ib_high}}^{(d)} = \delta_{\text{high}}^{(d)} \quad (22)$$

$$\mathcal{F}_{\text{ib_low}}^{(d)} = \delta_{\text{low}}^{(d)} \quad (23)$$

$$\mathcal{F}_{\text{prev_close_vs_ib}}^{(d)} = \frac{P_{\text{close}}^{(d)} - \text{IB}_{\text{mid}}^{(d)}}{\text{IB}_{\text{width}}^{(d)} + \varepsilon} \quad (24)$$

The third feature $\mathcal{F}_{\text{prev_close_vs_ib}}^{(d)}$ captures where the previous session's closing price settled relative to its own IB midpoint, encoding overnight directional commitment.

The conjunction of current spot position with historical IB levels yields cross-day confluence features. For example, `confluence_d1ibh_max_gamma` measures the geometric alignment between the prior day's IB high and the current day's dominant gamma strike:

$$\Phi_t^{(d1 \times \Gamma)} = \exp\left(-\frac{(\text{IB}_{\text{high}}^{(-1)} - K_{\Gamma}^*)^2}{2(b \cdot S_t)^2}\right) \quad (25)$$

High values of this feature indicate that a prior-day supply zone is coinciding with the current-day gamma pin, a particularly high-conviction resistance configuration.

6 Implied Volatility Surface Parameterization

6.1 ATM Implied Volatility and Term Structure

The ATM implied volatility σ_t^{ATM} for a given expiry T is the IV at the strike K closest to the current forward price $F_t = S_t e^{(r-q)(T-t)}$:

$$\sigma_t^{\text{ATM}}(T) = \text{IV}(K^*, T), \quad K^* = \arg \min_{K \in \mathcal{K}} |K - F_t| \quad (26)$$

The term structure of ATM IV captures the ratio of shorter to longer expiry volatilities:

$$\tau_{\text{struct}} = \frac{\sigma_t^{\text{ATM}}(\text{0DTE})}{\sigma_t^{\text{ATM}}(\text{weekly})} \quad (27)$$

A ratio $\tau_{\text{struct}} > 1$ (contango in volatility) indicates near-term elevated uncertainty; $\tau_{\text{struct}} < 1$ (backwardation) suggests calm short-term conditions.

6.2 Volatility Skew

The options market prices downside risk differently from upside risk. The risk-reversal skew at the 25-delta moneyness level quantifies this asymmetry:

$$\text{Skew}_{25\Delta} = \text{IV}(K_{25\Delta}^{\text{put}}) - \text{IV}(K_{25\Delta}^{\text{call}}) \quad (28)$$

High positive skew ($\text{Skew}_{25\Delta} \gg 0$) indicates elevated put demand, consistent with hedging pressure from large institutional long equity positions. Negative skew (“reverse skew”) is unusual in equity markets and may signal short-gamma crowding.

6.3 Smile Slope

The slope of the implied volatility smile along the strike axis provides a continuous measure of the skew at each expiry:

$$\text{SmileSlope}_{\text{call}} = \left. \frac{d\text{IV}}{dK} \right|_{K > F_t}, \quad \text{SmileSlope}_{\text{put}} = \left. \frac{d\text{IV}}{dK} \right|_{K < F_t} \quad (29)$$

These are estimated numerically from the discrete options chain via finite differences between the ATM and adjacent OTM strikes.

7 Cross-Ticker Feature Construction

7.1 VIX as a Volatility Regime Signal

The CBOE Volatility Index (VIX) measures 30-day implied volatility on the S&P 500. At 5-minute intervals, we compute:

$$x_t^{\text{VIX}} = \text{VIX}_t \quad (\text{level, not further transformed}) \quad (30)$$

$$r_{t,\Delta}^{\text{VIX}} = \frac{\text{VIX}_t - \text{VIX}_{t-\Delta}}{\text{VIX}_{t-\Delta}}, \quad \Delta \in \{1, 5\} \text{ min} \quad (31)$$

The raw VIX level is retained untransformed because the GBT can naturally partition on absolute VIX thresholds (e.g., “high regime: $\text{VIX} > 20$ ”), which are economically interpretable and stable across time.

7.2 TLT as a Risk-Off Signal

The iShares 20+ Year Treasury ETF (TLT) serves as a real-time proxy for flight-to-safety flows. Large positive TLT returns coinciding with negative SPX returns signal a risk-off environment where put demand is elevated:

$$\rho_{t,5}^{\text{SPX/TLT}} = \text{corr}\left(\{r_{s,1}^{\text{SPX}}\}_{s=t-4}^t, \{r_{s,1}^{\text{TLT}}\}_{s=t-4}^t\right) \quad (32)$$

This rolling 5-bar correlation is included to capture the real-time equity/bond correlation regime.

8 Temporal Feature Construction

The intraday session exhibits strong periodicity: liquidity, volatility, and option premium dynamics all follow systematic patterns across the 390-minute trading window. A naive `minutes_elapsed` feature would create splits that are specific to the numerical range $[0, 390]$ and would not generalize across the cyclical boundary between session close and next-day open.

Fourier basis encoding resolves this via:

$$x_{\text{tod}}^{\sin} = \sin\left(\frac{2\pi m_t}{390}\right), \quad x_{\text{tod}}^{\cos} = \cos\left(\frac{2\pi m_t}{390}\right) \quad (33)$$

where m_t is the number of minutes since 09:30. This places 09:30 and 16:00 at the same circular distance from the midday reference point, correctly encoding the symmetric time-decay acceleration at both market open and close.

Remark 1 (Interaction with Theta Decay). *For 0DTE options, theta decay is highly non-linear in time remaining. The raw `minutes_to_close` = $390 - m_t$ feature is retained alongside the Fourier features because it enables the GBT to learn threshold splits of the form “enter position only if $390 - m_t > 120$ ” (do not trade within 2 hours of expiry), which are not naturally expressible via the sinusoidal encoding.*

References

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